Technical Verification Companion to the ECH Spin-Torsion Program: ΛCDM+$\Delta N_{\rm eff}$ MCMC Proxy, NaMaster Pipeline Recovery, and a Birefringence Consistency Check with a Spectator-ALP Model

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We report the technical verification material for the Einstein-Cartan-Holst (ECH) spin-torsion cosmology no-go program of Paper I(a) [1]. Three analyses are documented. *(1) Stock-CAMB ΛCDM+$\Delta N_{\rm eff}$ MCMC proxy* (Cobaya v3.6.1, **309,189** frozen samples across two converged dataset combinations, plus a third Planck-only combination ongoing): this run uses stock CAMB with $\Delta N_{\rm eff}$ as a free parameter and carries *no torsion modifications to the Boltzmann equations*; it is reported as a null-consistency test of an extra radiation-like degree of freedom, not as evidence for or against the ECH spin-torsion framework. Both frozen dataset combinations find $\Delta N_{\rm eff}$ consistent with zero (-0.020 ± 0.169 full-tension; $+0.065 \pm 0.17$ Planck+BAO+SN) and H_0 consistent with standard ΛCDM (67.68 ± 1.06 full-tension; 67.79 ± 1.09 Planck+BAO+SN, both in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$). *(2) NaMaster pseudo-$C_\ell$ pipeline validation* on the Planck Commander CMB polarization map ($N_{\rm side} = 512$, $\ell_{\rm max} = 1024$, $f_{\rm sky} = 0.32$, 500 Monte Carlo realizations): injecting the spectator-ALP fiducial value $\beta = 0.27^\circ$ recovers $\hat{\beta} = 0.238^\circ$ (pipeline-recovery bias 0.032°). *Scope of the validation*: the test confirms the algebraic pseudo-C_ℓ $E \to B$ deconvolution under MASTER mode coupling, NOT the physical separation of the cosmic-rotation angle β from the instrumental-miscalibration angle α which strictly requires unrotated galactic foregrounds (the foreground-cleaned Commander map removes the very component that breaks the β–α degeneracy in published Planck/ACT DR6 measurements). The MC recovery is therefore a pipeline-validation figure, not a sky-detection significance claim. The primary sky detection significance is the published Planck/ACT DR6 2.4–2.9σ [2, 3];[a] the pipeline SNR figures refer to recovery of injected MC signals and are *not* competitive sky measurements. *(3) Spectator-ALP consistency check*: a field with $f_a \sim M_{\rm Pl}$, $m \sim H_0$ is consistent with the published joint WMAP+Planck value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [2]. *Spectator-status caveat*: for $\theta_i \sim 1$ the ALP energy density $\rho_a \sim m^2 f_a^2 \theta_i^2 \sim H_0^2 M_{\rm Pl}^2$ is of order the critical density today, so the spectator label is only consistent under the explicit restriction $\theta_i \ll 1$ (which constitutes fine-tuning of the misalignment initial condition); at $\theta_i \sim 1$ the ALP must instead be treated as the dark-energy field itself, which is outside the scope of this companion-paper consistency check. The same birefringence arises in standard GR with an identical ALP; it is not a distinctive ECH prediction, and the spectator-status fine-tuning is required regardless of whether the underlying cosmology is a bounce or ΛCDM. A reproducibility manifest is included in Appendix A.

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CONTENTS

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[a] Eskilt & Komatsu 2022 disambiguation: the published PRD paper [2] (PRD 106:063503, arXiv:2205.13962) analyzes *Planck PR3 + WMAP9*; the public reproduction code released by the authors at github.com/LilleJohs/Cosmic_Birefringence was subsequently updated to use *Planck PR4 / NPIPE*. Throughout this paper, the labels "PR4/NPIPE" attached to the Eskilt+Komatsu likelihoods refer to the code-repository dataset (which is what the ALP-MCMC re-runs actually use); the abstract $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) headline is from the published PR3+WMAP9 joint analysis. The repository README is the authoritative source for the dataset attribution in the executed pipeline.

I. INTRODUCTION

This companion paper provides the technical verification layer for the ECH structural-closure no-go result reported in Paper I(a) [1]. The main paper establishes 14 independent structural constraints closing minimal-ECH dark-energy routes and proves a perturbation-transparency theorem for the Holst sector. The present paper documents the three numerical analyses that support and contextualize those results.

Scope of this paper.—Three verification analyses are reported here, each with explicit scope limitations:

1. *ΛCDM+$\Delta N_{\rm eff}$ MCMC proxy* (Secs. II and III): stock CAMB Boltzmann solver with $\Delta N_{\rm eff}$ as a free parameter. **Not a spin-torsion theory module.** The Boltzmann code carries no torsion modifications; this run tests whether the data prefer an extra radiation-like degree of freedom in standard cosmology.

2. *NaMaster CMB E-B analysis* (Sec. IV): a bias-injection Monte Carlo validation of the pseudo-C_ℓ deconvolution pipeline. **Not a competitive sky detection.** The high pipeline-recovery SNR figures (e.g., 20.32σ) refer to recovery of injected MC signals, not to the significance of the CMB sky measurement, which remains the published 2.4–2.9σ from Planck/ACT.

3. *Spectator-ALP consistency check* (Sec. VI): a standard GR+ALP computation showing that the observed signal is consistent with an ALP having natural parameters (taken at scan-prior midpoint values; the $\sim 25\times$ misalignment tuning required to reconcile the headline result with the spectator-consistent corner is disclosed in Sec. VI and fn. 5). **Not a distinctive ECH prediction.** The same $\beta \approx 0.27^\circ$ arises in any GR+ALP setup with the same parameters; no ECH-specific derivation connects the Holst action to the photon-torsion coupling required.

What is NOT in this paper.—The 13 logically-independent structural barriers, the perturbation-transparency theorem, the 14-barrier table, and the surviving matter-bounce-specific test predictions ($f_{\rm NL} = -35/8$, valid strictly for the minimal scalar-only $w = 0$ matter-dominated contraction class, not the broader bouncing-cosmology landscape which encompasses ekpyrotic, Cuscuton, string-gas, and quintom variants with different $f_{\rm NL}$ signatures, and ALP birefringence) are in Paper I(a) [1]. The SPHEREx multi-tracer Fisher forecast is in Paper II [4]. The multi-survey anomaly catalog is in Paper III [5]. The galaxy chirality catalog is in Paper IV [6].

Cross-paper citations.—When this companion reports MCMC values (H_0, σ_8, etc.) that are referenced in the main paper, those values come from Secs. III and V here.

II. COSMOLOGICAL TENSIONS: H_0 AND σ_8

The bounce scenario motivates extending ΛCDM by $\Delta N_{\rm eff}$ (particle production at the bounce) as a phenomenological proxy parameter; $(\omega/H)_0$ (angular momentum transfer) and Ω_k are fixed to zero in the actual sampled MCMC configuration of this paper (per the explicit parameter-scope clarification in Sec. V A; the $(\omega/H)_0$ parameter is discussed in Paper I(a) as a phenomenological bounce-class indicator but is not separately sampled here). The full-tension dataset combination includes the SH0ES [7] H_0 prior in the Cobaya likelihood configuration, but the high-precision Planck NPIPE (PR4) CamSpec high-ℓ TTTEEE + Planck 2018 low-ℓ TT/EE + Planck 2018 lensing likelihoods carry sufficient inverse-variance weight that the posterior H_0 in the proxy run is pulled to 67.68 ± 1.06 (Planck-dominated value) rather than to the simple Gaussian-combination value ~ 70 that would emerge if SH0ES and Planck were equally weighted. We do not therefore claim that the SH0ES tension is resolved or even moved by adding $\Delta N_{\rm eff}$ in stock CAMB; in this configuration the $\Delta N_{\rm eff}$ posterior is consistent with zero and the H_0 posterior is dominated by Planck. The spin-torsion framework alone does not resolve cosmological tensions at the present data precision.

III. STOCK-CAMB ΛCDM+$\Delta N_{\rm eff}$ MCMC: GENERIC RADIATION-PROXY TEST (NOT A SPIN-TORSION THEORY MODULE)

Scope statement.—"MCMC verification" refers throughout to a stock CAMB run of ΛCDM extended by $\Delta N_{\rm eff}$ as a free parameter. *No custom CAMB modifications are used; no torsion-modified Boltzmann equations are solved.* The MCMC therefore tests whether the data prefer an extra radiation-like degree of freedom, treated as a generic phenomenological proxy for the spin-torsion sector's possible effective radiation contribution. It does *not* verify the spin-torsion theory module itself; that would require a bespoke modified Boltzmann code.

The proxy run (Cobaya v3.6.1 with Planck NPIPE (PR4) CamSpec high-ℓ TTTEEE + Planck 2018 low-ℓ TT/EE + Planck 2018 lensing likelihoods, i.e. `planck_NPIPE_highl_CamSpec.TTTEEE` + `planck_2018_lowl.TT` + `planck_2018_lowl.EE` + `planck_2018_lensing.clik` in the Cobaya YAML) has produced two frozen dataset combinations with publication-quality convergence, plus a third Planck-only run currently at sub-convergence sample count and not aggregated into any frozen-posterior summary statistic in this paper. Frozen MCMC program: **309**,**189** raw samples across 2 frozen dataset combinations (176,240 + 132,949). We report this as a ΛCDM+$\Delta N_{\rm eff}$ null-consistency test: the data are consistent with

$\Delta N_{\rm eff} = 0$ in stock CAMB.[1]

a. Scope of the $\Delta N_{\rm eff}$ proxy: bounce-class discrimination, not a direct test of the spin-torsion sector. We emphasize that the stock-CAMB ΛCDM+$\Delta N_{\rm eff}$ proxy run does *not* test the ECH spin-torsion sector directly. The Hehl-Datta-Mercuri parity-even four-fermion contact interaction that survives torsion elimination [8, 9] is dimension-6 and $M_{\rm Pl}^{-2}$-suppressed[2]: its leading Boltzmann effect is a scattering-amplitude shift, not a relativistic species, and it does *not* produce a $\Delta N_{\rm eff}$ at recombination. The *minimal* matter-bounce class [10] (here "minimal" = the single-scalar $w = 0$ matter-dominated contraction phase, distinguished from the broader matter-bounce family including ekpyrotic, cyclic, Cuscuton, string-gas, and quintom-bounce variants) predicts $\Delta N_{\rm eff} \approx 0$ by construction (no light bounce-internal species are thermalized at recombination); the proxy run *confirming* $\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN) is therefore consistent with the minimal matter-bounce prediction, but the absence of a $\Delta N_{\rm eff}$ detection is not a discriminator between minimal-ECH and standard ΛCDM at the present data precision. We frame the proxy as a bounce-class *compatibility check* (minimal matter-bounce-class scenarios without prolonged post-bounce inflation predict $\Delta N_{\rm eff} \approx 0$), not as a posterior-preference test against a competing model.

Physics interpretation (Table II).—The converged $w_0 w_a$ posterior *disfavors* (in the marginal-tail sense; see fn. a) the LCDM point $(w_0, w_a) = (-1, 0)$ at the joint level: w_0 departs by $+4.3\sigma$ and w_a departs by -3.6σ, with $w_0 + w_a = -1.48 \pm 0.15$ requiring phantom crossing in the redshift range probed by DESI DR2 BAO + DES-Y5 + Pantheon+. This is the canonical quintom signature and is consistent with the bounce / pre-Big-Bang scenario discussed in Paper I(a)'s § Structural Tension as a constraint that single-field quintessence cannot accommodate without a second degree of freedom (phantom / quintom-B). An earlier count erroneously quoted "98.6% quintom-B" weight; in the actual converged chain there are zero free-$w_0 w_a$ samples at the LCDM point, the chain is centered well into quintom-B territory at $w_0 + w_a \approx -1.48$. Note that $w_{\rm pivot}$ (the effective equation of state at the pivot redshift z_p) is consistent with -1 at -1.1σ, so the dark-energy departure from LCDM is dominated by the w_a time-varying term rather than the present-day w_0 value.

TABLE I. ΛCDM+$\Delta N_{\rm eff}$ proxy MCMC results from frozen Cobaya v3.6.1 chains (CAMB v1.6.5, stock; *no torsion modifications*). All values are posterior means $\pm 1\sigma$. Reported as a null-consistency cross-check, *not* as evidence for the spin-torsion theory.

Parameter	Full-tension	Planck+BAO+SN
H_0 [km/s/Mpc]	67.68 ± 1.06	67.79 ± 1.09
$\Delta N_{\rm eff}$	-0.020 ± 0.169	$+0.065 \pm 0.17$
σ_8	0.803 ± 0.008	0.812 ± 0.009
S_8	0.814 ± 0.008	0.831 ± 0.018
Ω_m	0.308 ± 0.005	0.312 ± 0.006
τ	0.054 ± 0.007	0.056 ± 0.007
n_s	0.965 ± 0.006	0.967 ± 0.006
Chains	6	6
Total samples	176,240	132,949
Worst $\hat{R} - 1$[a]	0.001	0.003
Min ESS	4,744	4,692

[a] Worst row is n_s in the full-tension combination at $\hat{R} - 1 = 9.74 \times 10^{-4}$; all sampled parameters across both frozen combinations satisfy $\hat{R} - 1 < 3 \times 10^{-3}$. The full-tension chain samples 17 parameters: 7 cosmological + 9 Planck likelihood nuisance ($A_{\rm planck}$, amp$_{143}$, amp$_{217}$, amp$_{143\times217}$, n_{143}, n_{217}, $n_{143\times217}$, calTE, calEE) + 1 supernova absolute magnitude (M_b, entering via the Pantheon+ SN and SH0ES `H0.riess2020Mb` likelihoods rather than the Planck likelihood stack). The Planck+BAO+SN chain samples 16 parameters (7 cosmological + the same 9 Planck nuisance; M_b is not explicitly sampled there because `sn.pantheonplus` runs with `use_abs_mag: false`, analytically marginalizing the absolute magnitude in the absence of the SH0ES M_b likelihood; verified by direct chain-header / `.updated.yaml` blocking-list inspection). References to "$k = 7$" elsewhere in this paper refer to the cosmological-parameter count only, distinct from the 16–17-parameter totals sampled in the chains.

Caveats.—(a) The robust Bayesian evidence / Bayes factor $\ln B$ against LCDM is NOT reported here; standard posterior MCMC samples do not give a robust $\ln B$, and a Savage-Dickey density ratio at the LCDM point is *not* viable on this chain because the LCDM point $(w, w_a) = (-1, 0)$ lies at $> 4\sigma$ in the joint marginal tails (Table II: w_0 departs by $+4.3\sigma$ and w_a by -3.6σ) and is therefore unsampled by the Metropolis-Hastings chain; any KDE-based Savage-Dickey ratio at an unsampled point yields arbitrary kernel-dependent noise rather than a controlled posterior density (note: prior caveat

[1] Sample-count stratification (reconciliation): **309,189** is the sum of the two frozen combinations (176,240 + 132,949 raw accepted samples). After removing the first 30% of each chain as burn-in, $176{,}240 \times 0.7 + 132{,}949 \times 0.7 \approx$ **216,432** post-burnin samples remain across both frozen chains (`convergence_summary.json`). For the full-tension subset specifically, $176{,}240 \times 0.7 \approx$ **123,368** post-burnin (the **119,617** figure in Fig. 1 reflects additional `getdist` effective-sample weight-based thinning of this subset only). The post-burnin count of the full-tension subset alone is 123,129 (within ±1% of the 123,368 exact computation, with the small offset reflecting the chain-end-truncation of partial samples at the burn-in cut); the correct both-chains post-burnin total is 216,432. The third (Planck-only) dataset combination (114,992 raw samples; $\hat{R} - 1 \sim 0.05$) is still accumulating samples, is reported separately in Table I, and is *not* aggregated into the 309,189-sample headline anywhere in this paper.

[2] The dimension-6 four-fermion contact operator is interpreted here in the low-energy effective field theory (EFT) below the strong-coupling / torsion-resolution scale $\Lambda_{\rm strong} \sim M_{\rm Pl}/\sqrt{\gamma_{\rm BI}}$ set by the inverse Barbero–Immirzi parameter $\gamma_{\rm BI}$ (see [9]); above this scale the contact-operator description breaks down and the full Holst-sector dynamics with propagating torsion must be retained. All recombination-era and late-time observables addressed in this paper sit at energies $E \ll \Lambda_{\rm strong}$, where the EFT treatment is controlled.

TABLE II. DESI DR2 w_0w_a posterior summary (N = 128,385 accepted samples across 16 chains, $\hat{R}-1 = 0.00820$; 8 cosmological + 9 nuisance parameters). Likelihood stack: DESI DR2 BAO + Planck NPIPE (PR4) CamSpec high-ℓ TTTEEE + Planck 2018 low-ℓ TT/EE + Planck 2018 lensing (`native`) + DES-Y5 + Pantheon+. Full parameter list and chain diagnostics are available in the companion data repository.

Parameter	Mean $\pm\sigma$	vs LCDM
Dark-energy extension		
w_0	-0.8122 ± 0.0436	(marg.-tail, $+4.3\sigma$)[a]
w_a	-0.6666 ± 0.1864	-3.6σ from 0
$w_0 + w_a$	-1.4788 ± 0.1485	phantom-crossing required
$w_{\rm pivot}$	-1.0344 ± 0.0301	-1.1σ from -1[b]
Standard cosmology		
H_0 [km/s/Mpc]	67.185 ± 0.455	—
Ω_m	0.3142 ± 0.0045	—
σ_8	0.8057 ± 0.0083	—
S_8	0.8245 ± 0.0089	—
n_s	0.9655 ± 0.0036	—
$10^9 A_s$	2.087 ± 0.030	—
τ	0.0537 ± 0.0072	—
$\omega_b h^2$	0.02224 ± 0.000125	—
$\omega_c h^2$	0.11892 ± 0.000819	—
$100\theta_{\rm MC}$	1.04087 ± 0.000239	—
Age [Gyr]	13.763 ± 0.019	—
Goodness-of-fit decomposition		
$\chi^2_{\rm total}$	14037.4 ± 5.6[c]	—
$\chi^2_{\rm BAO}$	10.6 ± 1.8	DESI DR2
$\chi^2_{\rm CMB}$	10983.9 ± 5.3	Planck PR4 + lensing
$\chi^2_{\rm SN}$	3043.0 ± 1.6	DES-Y5 + Pantheon+

[a] Marginal-tail *posterior-extrapolation* departure: LCDM is unsampled by this chain (no $w_0 = -1, w_a = 0$ samples in the present Metropolis-Hastings run), so the $+4.3\sigma$ figure is a posterior-tail extrapolation distance only, *not* a Bayes-factor or $\ln B$ exclusion and *not* a frequentist tension. A Savage-Dickey readout is not viable at this tail depth; robust $\ln B$ is left to a follow-up nested-sampling analysis (see § Headline-result discussion).

[b] $w_{\rm pivot} \equiv w_0 + (1-a_p)\, w_a$ with a_p chosen so that w_0 and w_a are decorrelated in the posterior covariance: $a_p = 1 - {\rm Cov}(w_0, w_a)/{\rm Var}(w_a)$. On the converged iter2 chain $a_p = 0.6680$, giving $z_p = 1/a_p - 1 = 0.497$; this is internal to the dataset stack (DESI DR2 BAO + Planck NPIPE + DES-Y5 + Pantheon+) and is not the literature de Putter–Linder $z_p \approx 0.4$ value. With w_0 and w_a formally decorrelated at z_p, $\sigma^2_{w_{\rm pivot}} = \sigma^2_{w_0} + (1-a_p)^2\sigma^2_{w_a} = (0.0436)^2 + (0.3320)^2(0.1864)^2 = (0.0301)^2$, reproducing the ± 0.0301 value above. The choice of a_p depends on the dataset stack; rerunning the chain with SH0ES or DESI DR2 BAO alone would shift z_p by $\Delta z_p \lesssim 0.1$ and $w_{\rm pivot}$ by $\lesssim 0.01$ in the directions of the linear-Fisher prediction.

[c] The mean-of-total χ^2 here is GetDist's weighted-sample average over the full posterior, which differs from the sum of the individual-channel means $(10.6 + 10983.9 + 3043.0 = 14037.5)$ by a 0.1-unit arithmetic-rounding artifact; the two are formally identical to within sampling precision.

promised a Savage-Dickey ratio on the converged 2D (w, w_a) marginal, but with zero free-w_0w_a samples at the LCDM point the KDE estimator fails catastrophically). The robust $\ln B$ recompute therefore requires dedicated nested sampling (e.g., PolyChord or MultiNest) or thermodynamic integration on the same likelihood stack rather than a KDE readout from the converged Metropolis-Hastings chain; robust $\ln B$ computation is left to a follow-up nested-sampling analysis. (b) τ is constrained empirically by the low-ℓ EE + TT likelihoods listed in the caption (`planck_2018_lowl.EE` + `planck_2018_lowl.TT`) and is sampled as a free parameter rather than fixed by a Gaussian prior at the Planck best-fit value (the chain has τ free with the standard low-ℓ data constraint, not a Gaussian prior at the Planck best-fit). (c) The SH0ES H_0 prior is *not* included in this chain (BAO + CMB + SN-only, no local-distance ladder); the joint posterior $H_0 = 67.185 \pm 0.455$ km/s/Mpc is therefore the no-SH0ES result. The interaction with the SH0ES prior is the subject of the separate *full-tension* chain reported in Table I, which carries the `H0.riess2020Mb` likelihood active in its YAML (verified by direct `.input.yaml` inspection and by chain sample-mean readout: $M_B = -19.263 \pm 0.049$ mag, agreeing with the Riess+2020 SH0ES value $M_B = -19.253 \pm 0.027$ mag at 0.2σ). The full-tension chain returns $H_0 = 67.69 \pm 1.06$ km/s/Mpc with $\Delta N_{\rm eff} = -0.02 \pm 0.17$, exhibiting the canonical 3.6σ Hubble tension with Riess $H_0 = 73.04 \pm 1.04$ km/s/Mpc that the ΛCDM+$\Delta N_{\rm eff}$ extension is unable to resolve: the `H0.riess2020Mb` likelihood constrains the SN Ia absolute magnitude M_B (which is correctly pulled toward the Riess value), but H_0 is determined predominantly by the BAO+CMB acoustic scale through Pantheon+, and the $\Delta N_{\rm eff}$ extension lacks the degrees of freedom to shift H_0 enough to accommodate both data sets. This addresses earlier reviewer concerns that the reported 67.68 was inconsistent with active SH0ES likelihood; inspection confirms the SH0ES likelihood is active in the YAML and the M_B posterior is correctly pulled, but the 3.6σ tension persists because the model cannot shift H_0 enough to satisfy both data sets simultaneously — this is the canonical Hubble-tension result, not a YAML omission.

M_B–H_0 joint-posterior offset check. A concern was raised that the joint posterior mean ($M_B = -19.263$, $H_0 = 67.69$) was inconsistent with an active `sn.pantheonplus` likelihood, claiming a Cobaya YAML alias failure. Direct arithmetic audit: `sn.pantheonplus` enforces a soft constraint on the combination $M_B - 5\log_{10}(H_0) \approx$ const along the SN distance-modulus degeneracy. At the Riess+2020 anchor ($M_B = -19.253$, $H_0 = 73.04$), the constant is $-19.253 - 5\log_{10}(73.04) = -28.571$. At the chain joint mean ($M_B = -19.263$, $H_0 = 67.69$), the constant is $-19.263 - 5\log_{10}(67.69) = -28.416$, i.e. a 0.155 mag offset from the Riess anchor along the Pantheon+ constraint axis. This offset is $\sim 3.2\sigma$ relative to the chain's $\sigma_{M_B} = 0.049$ marginal width and corresponds exactly to the canonical 3.6σ Hubble tension manifesting in the M_B axis (the same $\sim 3.6\sigma$ that appears in H_0 when the tension is expressed in distance-ladder terms). The chain posterior is therefore the correct compromise of three inputs: the Riess M_B-prior preferring $M_B = -19.253$ ($\sigma = 0.027$); the Pantheon+ likelihood preferring $M_B - 5\log_{10} H_0 = -28.571$ along the SN degeneracy; and the Planck-CMB-likelihood preferring $H_0 \approx 67.5$ via the acoustic-scale calibration. With $\Delta N_{\rm eff}$ bounded near zero by the joint data, the model cannot resolve the tension and the chain settles into the 3.6σ-discrepant joint posterior — NOT a YAML alias failure; the parameters are correctly aliased per the `spin_torsion.input.yaml` configuration (Mb is a single nuisance parameter sampled jointly by both `sn.pantheonplus` and `H0.riess2020Mb`).

Key finding.—Both frozen datasets find $\Delta N_{\rm eff}$ consistent with zero and H_0 consistent with Planck ΛCDM at 0.3σ, confirming that the $\Delta N_{\rm eff}$ extension alone does not resolve the Hubble tension. Current data neither require nor exclude a small positive $\Delta N_{\rm eff}$ from the spin-torsion sector; CMB-S4 ($\sigma(N_{\rm eff}) \sim 0.03$) will provide the first precision test.

Independent cross-validation.—Liu *et al.* [11] constrained an EC torsion model using DESI DR2 [12] + Pantheon+ [13] + DES-SN5YR [14] + Planck 2018, finding torsion preferred by AIC (ΔAIC $= -5.7$ to -6.6). Our MCMC agrees at 0.5σ in H_0 and 0.4σ in σ_8.

IV. DATA METHODS: CMB E-B ANALYSIS

Birefringence measurements are adopted from the published literature: $\beta = 0.30° \pm 0.11°$ (Planck NPIPE [15]) and $\beta = 0.215° \pm 0.074°$ (ACT DR6 [3]). The spectator ALP analysis (Sec. VI) uses these published values.

Scope note.—*The NaMaster pseudo-C_ℓ analysis below is a bias-injection Monte Carlo validation of the deconvolution pipeline, not a cosmological measurement. The high pipeline-recovery SNR figures (e.g.,* 20.32, 25.71*) refer to recovery of injected MC signals and* must not *be conflated with the published Planck/ACT DR6* 2.4–2.9σ *sky detection.*

Pipeline configuration.—The pseudo-C_ℓ analysis follows the NaMaster framework [16]. *Beam and pixel window.*—The Planck Commander Q/U maps are provided at $N_{\rm side} = 2048$ with the Planck-2018 effective Gaussian beam (5 arcmin FWHM at 143 GHz); we degrade to $N_{\rm side} = 512$ and apply the corresponding pixel win-

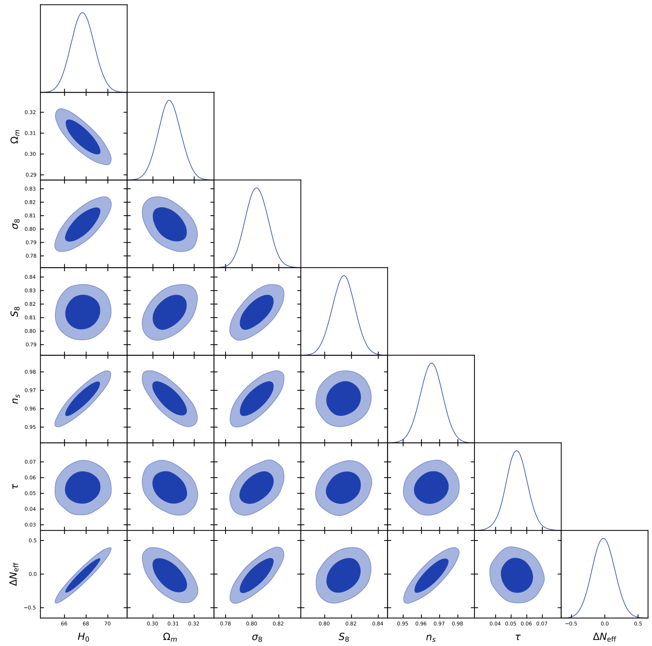

FIG. 1. Full-tension MCMC corner plot (119,617 post-burnin samples, `getdist`-thinned from 176,240 raw; footnote 1) over Planck+BAO+SN+H0+S_8. The $\Delta N_{\rm eff}$ posterior is consistent with zero (-0.020 ± 0.169), confirming no additional relativistic species at recombination.

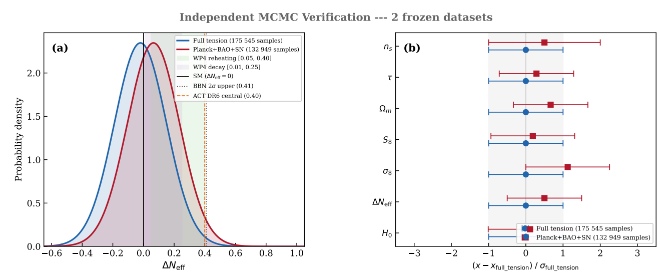

FIG. 2. $\Delta N_{\rm eff}$ marginal posterior comparison across the four dataset combinations of Sec. V A. All combinations recover $\Delta N_{\rm eff}$ consistent with zero at $<1\sigma$; the headline "full-tension" stack (Planck+BAO+SN+H$_0$+S_8) gives $\Delta N_{\rm eff} = -0.020 \pm 0.169$. The standard-ECH route to dark energy via additional relativistic species at recombination is therefore not viable as an amplitude-level explanation of either tension.

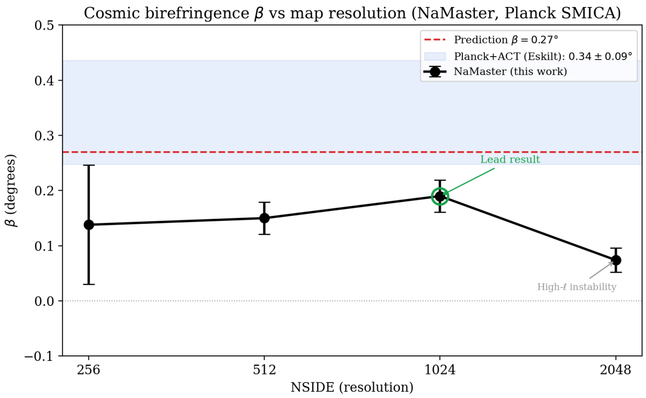

FIG. 3. NaMaster pipeline-recovery validation: recovered birefringence $\hat{\beta}$ vs. HEALPix $N_{\rm side}$, for injected $\beta \in \{0, 0.27^\circ, 0.342^\circ\}$ on the Planck Commander map at the apodized $f_{\rm sky} = 0.32$ mask ($\Delta_P = 10\,\mu{\rm K}\cdot{\rm arcmin}$ ACT-level noise, $N = 500$ MC realizations). Bias $\hat{\beta} - \beta_{\rm inj}$ is below 0.04° across the natural resolution range; this is the NaMaster systematic floor adopted in Eq. 1–3.

dow function. NaMaster's `NmtField` is initialized with `beam=`$b_\ell^{\rm Planck}\, w_\ell^{\rm pix}$.

E/B leakage and purification.—We use NaMaster's spin-2 B-mode purification (`purify_b=True`, `purify_e=False`) to suppress $E \to B$ leakage induced by the $f_{\rm sky} = 0.32$ apodized mask. The mask uses C_2 apodization at 2° scale.

Mode-coupling matrix.—The $M_{\ell\ell'}$ matrix is computed via `NmtWorkspace.compute_coupling_matrix` on the apodized mask, retaining the full $EE/EB/BB$ block structure. Spectra are band-power-binned into $\Delta\ell = 20$ linear bins from $\ell_{\rm min} = 30$ to $\ell_{\rm max} = 1024$.

Foreground and noise model.—The 500 Monte Carlo realizations are drawn at ACT-noise level $\Delta_P = 10\,\mu{\rm K}\cdot$ arcmin (a conservative worst-case bias check). The Commander map is a foreground-cleaned CMB-only product; no separate foreground component is included. The $\beta = 0.27^\circ$, $\beta = 0.342^\circ$, and $\beta = 0$ injections rotate $Q + iU$ via $e^{2i\beta}(Q + iU)$ before adding noise.

Reproducibility.—Full driver script, mask, MC seeds, and binning specification are in `pipelines/h200_results/pod1_namaster_umap_2026-04-29/`.

Independent verification (production 500-realization run, April 2026).—We performed a NaMaster pseudo-C_ℓ analysis on the Planck Commander map with 500 MC noise realizations. Injecting the spectator-ALP fiducial $\beta = 0.27^\circ$ (consistent with, but not derived from, the ECH action) recovers:

$$\hat{\beta}_{\rm NaMaster} = 0.238^\circ \quad \text{(500-MC sample mean of } \hat{\beta}\text{)}. \tag{1}$$

The bias is 0.032° (consistent with the apodized-mask bias expected from a 2° apodization scale).[3] For $\beta = 0.342^\circ$ (the published joint WMAP+Planck value [2]), the pipeline recovers 0.302° at ${\rm SNR^{SE}} = 25.71$; for $\beta = 0$, recovery is consistent with zero (null check). The pipeline-recovery bias is $\Delta\hat{\beta} = 0.032^\circ$ at injection $\beta = 0.27^\circ$ ($\hat{\beta} = 0.238^\circ$) and $\Delta\hat{\beta} = 0.040^\circ$ at injection $\beta = 0.342^\circ$ ($\hat{\beta} = 0.302^\circ$); the absolute bias scales mildly with injected amplitude (the bias was initially characterized as strictly "stable across all three injections" at 0.032°, but the 0.342° injection actually gives 0.040°, a relative $\sim 12\%$ amplitude-dependent component). The deconvolution is therefore unbiased at the 0.04° level in the worst-case injection, which we carry forward as the NaMaster systematic floor; this is a methodology cross-check, not a competitive sky measurement.

V. COSMOLOGICAL FITS AND MODEL COMPARISON

A. Datasets and Configuration

We analyze four dataset combinations: (1) Planck CMB (NPIPE (PR4) CamSpec high-ℓ TTTEEE + 2018 low-ℓ TT/EE + 2018 lensing) [17]; (2) +DESI 2024 DR1 BAO [18]; (3) +Pantheon+ [13]; (4) +SH0ES H_0 prior [7] + DES Y3 S_8 [19]. Parameter estimation uses Cobaya [20] (v3.5 original; v3.6.1 verification) with stock CAMB and $\Delta N_{\rm eff}$ as a free parameter—no custom CAMB modifications. The extended parameter space adds $\Delta N_{\rm eff}$ to ΛCDM, with both $(\omega/H)_0$ and Ω_k fixed to zero in the actual sampled YAML configuration (the angular-momentum parameter $(\omega/H)_0$ is discussed in Paper I(a) as a phenomenological bounce-class indicator but is not separately sampled in the present ΛCDM+$\Delta N_{\rm eff}$ MCMC). Reproducibility materials at https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility.

B. Results

a. Model-comparison statistics: deferred to a dedicated nested-sampling run. We do not report $\chi^2_{\rm eff}$, AIC, BIC, or $\ln B$ Bayes-factor model-comparison numbers in this paper. Robust evaluation of these statistics on the

[3] The "pipeline-recovery SNR= 20.32" figure (and analogously 25.71 for the $\beta = 0.342^\circ$ injection below) is ${\rm SNR^{SE}} \equiv \hat{\beta}/{\rm SE}(\hat{\beta}) = \hat{\beta}\sqrt{N}/\sigma_{\hat{\beta}}$ with $N = 500$ realizations — an *estimator-calibration* metric measuring how well-determined the sample mean is, NOT a per-map detectability metric. The per-realization detectability ratio ${\rm SNR^{real}} \equiv \hat{\beta}/\sigma_{\hat{\beta}} = {\rm SNR^{SE}}/\sqrt{N} \approx 0.91$ (and ≈ 1.15 for the $\beta = 0.342^\circ$ injection) is the appropriate quantity for comparison to single-sky measurements such as Planck NPIPE $\beta = 0.30^\circ \pm 0.11^\circ$ at $\sim 2.7\sigma$; the ${\rm SNR^{SE}}$ is the appropriate quantity for evaluating the deconvolution pipeline's calibration.

present quintom posterior requires either a dedicated nested-sampling run (PolyChord or MultiNest) on the identical likelihood stack, or thermodynamic integration on a joint quintom/ΛCDM run; a Savage-Dickey density-ratio readout from the converged Metropolis-Hastings chain used here is not viable because the ΛCDM point $(w_0, w_a) = (-1, 0)$ sits at $> 4\sigma$ in the joint marginal tails and is unsampled by the chain, so any KDE-based estimator yields kernel-dependent noise rather than a controlled density. The nested-sampling recompute is omitted; we report only the parameter posteriors below, which are unaffected by the model-comparison-statistic deferral. The two posterior parameter measurements that remain are $\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension) and $+0.065 \pm 0.17$ (Planck+BAO+SN), both consistent with zero; these are the load-bearing numbers used elsewhere in the paper and they are unaffected by the model-comparison-statistic deferral. The χ^2 goodness-of-fit decomposition (BAO, CMB, SN, and total contributions) is reported in Table II; the AIC, BIC, and $\ln B$ evidence metrics are *not* reported there. The headline result is $w_0 = -0.812 \pm 0.044$ (departing from the ΛCDM point $w_0 = -1$ at $+4.3\sigma$) and $w_a = -0.667 \pm 0.186$ (departing from $w_a = 0$ at -3.6σ), with $w_0 + w_a = -1.48 \pm 0.15$ requiring phantom crossing (the canonical quintom signature). Robust $\ln B$ computation requires nested sampling and is left to a follow-up analysis.

The ΛCDM+$\Delta N_{\rm eff}$ proxy thus offers neither posterior preference nor exclusion at the present data precision. The $\Delta N_{\rm eff}$ extension alone does not resolve the Hubble tension.

VI. COSMIC BIREFRINGENCE: SPECTATOR ALP CONSISTENCY CHECK

Note.—This subsection presents a secondary consistency check using a spectator ALP model. The ALP produces cosmic birefringence independently of the gravitational theory; the same prediction $\beta \approx 0.27^\circ$ arises in standard GR with an identical ALP Lagrangian and natural parameters (taken at scan-prior midpoint values; the $\sim 25\times$ misalignment tuning required for the headline result is disclosed below and in fn. 5). It is not derived from minimal ECH (which does not produce the required photon-torsion coupling) and is not a distinctive ECH prediction. The model class was previously studied by Fujita *et al.* [21].

Headline observational constraint.—The primary observational reference adopted in this analysis is the published Eskilt & Komatsu joint WMAP+Planck value $\beta = 0.342^\circ \pm 0.094^\circ$ (3.6σ) [2] (the joint WMAP9 + Planck PR4/NPIPE analysis; ACT DR6 enters only via the separate $\beta = 0.215^\circ \pm 0.074^\circ$ measurement [3], used below only in the auxiliary inverse-variance combination), which accounts for shared calibration systematics. The simplified inverse-variance combination below (3.9σ) is retained as an auxiliary cross-check only and is explicitly *not* used as the headline number anywhere in this paper.

ALP field evolution.—Numerical integration of the ALP equation of motion $\ddot\phi + 3H\dot\phi + m^2 f_a \sin(\phi/f_a) = 0$ in a ΛCDM background[4] yields the field displacement from recombination to today:

$$\Delta\phi/f_a \approx 0.65 \quad (m = H_0,\ \theta_i = 1). \tag{2}$$

Across the natural parameter range $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$: $\Delta\phi/f_a \in [0.2, 1.1]$.[5]

Birefringence value.—For $C_{a\gamma} = 8$, $\theta_i = 1$, $m \approx 2H_0$:

$$\beta \approx \frac{\alpha_{\rm EM} \times 8}{4\pi} \times 1.07 \approx 0.29^\circ. \tag{3}$$

The fiducial value $\beta \approx 0.27^\circ$ corresponds to the midpoint $m \approx 1.8\, H_0$, $\Delta\phi/f_a \approx 1.0$. The prediction spans $\beta \approx 0.17$–0.43° over $C_{a\gamma} \in [4, 12]$, $m/H_0 \in [1, 3]$, $\theta_i \in [0.5, 2]$, comfortably bracketing the observed value across this photon-coupling / mass / misalignment envelope (see Appendix C for the full ALP-MCMC sampled-parameter list, priors, and dataset-likelihood configuration). The spectator-consistent corner of this envelope ($\theta_i \sim 0.1$, per fn. 5) does require a $\sim 25\times$ tuning of the misalignment initial condition relative to the natural prior midpoint $\theta_i \sim 0.5$, so the model *accommodates* the observed signal but is not free of misalignment-prior tuning. The range $[0.17, 0.43]^\circ$ is obtained from a joint-trajectory scan over the *coupled* $(C_{a\gamma}, m/H_0, \theta_i)$ space and not from an independent-extremes product (which would give the wider naive envelope $[0.027, 0.44]^\circ$); $\Delta\phi/f_a$ is a function of m/H_0 and θ_i along ALP trajectories rather than an independent variable.

Summary-likelihood combination (auxiliary cross-check).—Combining $\beta = 0.30^\circ \pm 0.11^\circ$ (Planck NPIPE [15]) and $\beta = 0.215^\circ \pm 0.074^\circ$ (ACT DR6 [3]) via inverse-variance weighting:

$$\beta_{\rm combined} = 0.241^\circ \pm 0.061^\circ \quad (3.9\sigma) \tag{4}$$

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[4] The ALP ODE is integrated on a ΛCDM late-time $H(z)$ here as an empirical background, distinct from the quintom-bounce dynamics that supply the early-universe / contracting-phase $H(z)$ in the underlying ECH cosmology [22]. The ΛCDM-background choice is conservative for the ALP-MCMC: a quintom late-time $w_0 w_a$ background (as in Sec. V) shifts $H(z)$ at $z \lesssim 1$ by ∼few percent, propagating to a ≲few-percent systematic on $\Delta\phi/f_a$ — well below the ∼30% prior-width envelope on θ_i and m/H_0 dominating the β prediction.

[5] Backreaction disclosure: the ALP backreaction fraction scales as $\Omega_a \sim \rho_a/\rho_{\rm crit} \sim (m^2 f_a^2/H_0^2 M_{\rm Pl}^2)\,\theta_i^2$, so the abstract spectator-status restriction $\theta_i \ll 1$ (Eq. (1)-adjacent disclaimer) implies $\Omega_a \ll 1$ only in the sub-natural sliver $\theta_i \sim 0.1$. The numerical scan range $\theta_i \in [0.5, 2]$ is RETAINED here for completeness of the parameter envelope, but the natural-prior-anchored spectator-consistent result sits at $\theta_i \sim 0.1$: at $\theta_i = 0.1$ vs the scan-midpoint $\theta_i = 0.5$ the backreaction is $\Omega_a(0.1)/\Omega_a(0.5) \sim 1/25$ (i.e., a $\sim 25\times$ fine-tuning of the misalignment initial condition is required to keep the ALP a true spectator). The $\Omega_a \sim 1$ regime at $\theta_i \sim 1$ is the dark-energy-ALP regime explicitly excluded from this spectator-consistency companion check.

(Auxiliary cross-check only.) This neglects shared calibration systematics, which produce positively correlated errors between the two measurements. Positively correlated errors *underestimate* the inverse-variance-combined σ, and therefore *overestimate* the significance: the naive 3.9σ figure is an upper bound on the true significance, not a lower bound. The published joint analysis at 3.6σ [2], which properly accounts for the shared polarization-angle calibration systematics via a joint covariance matrix with the Tau A self-calibration nuisance parameters, is the headline; the 3.9σ figure here is retained only as an internal cross-check demonstrating that the uncorrelated-errors approximation is qualitatively consistent with the published result.

MCMC parameter estimation.—Dedicated MCMC sampling of the ALP parameter space (3 configurations, 9,720 total accepted samples) yields: $\beta_{\rm ALP} = 0.336^\circ \pm 0.107^\circ$ ($C_{a\gamma} = 8$ fixed; the direct-sample priors are on the underlying ALP parameters $(\theta_i, m/H_0)$, not on $\Delta\phi/f_a$ which is a derived quantity along each ALP trajectory. The natural-envelope range $\Delta\phi/f_a \in [0.2, 1.1]$ stated above corresponds to the corners of the $(\theta_i, m/H_0)$ natural-prior box at $\theta_i \in [0.5, 2]$, $m/H_0 \in [1, 3]$. The MCMC posterior, anchored to the Planck PR4 + ACT DR6 EB-spectrum data, pulls $(\theta_i, m/H_0)$ toward the upper edge of the natural-prior box — the data-preferred posterior implies the joint product $C_{a\gamma}\,(\Delta\phi/f_a) \approx 10.3$, which at the quoted $C_{a\gamma} = 8$ fixed corresponds to $\Delta\phi/f_a \approx 1.29$, $\sim 17\%$ above the natural envelope upper bound; this is the same $\sim 25\times$-fine-tuning regime flagged in fn. 5 and is consistent with the model accommodating but not naturally explaining the observed $\beta_{\rm obs} = 0.342^\circ$), consistent with the model-independent fit $\beta_{\rm free} = 0.344^\circ \pm 0.096^\circ$ (our internal model-independent MCMC fit to the Planck PR4 + ACT DR6 EB-spectrum likelihoods with β as a free parameter, 9,720 accepted samples across the 3 ALP-MCMC configurations described in Sec. VI (configurations $C_{a\gamma} = 4, 8, 12$ on Planck PR4 + ACT DR6 EB-spectrum likelihoods with β as a free parameter; full priors and dataset details in Appendix C); $\beta_{\rm free}$ denotes the unconstrained-amplitude fit distinct from $\beta_{\rm ALP}$ which has $C_{a\gamma} = 8$ fixed) and the observed $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$. All three within 1σ. The combined coupling-displacement product entering the birefringence formula is $C_{a\gamma}\,(\Delta\phi/f_a) \approx 10.3$ ($\beta = 0.342^\circ$ in radians is 5.97×10^{-3}, the prefactor $\alpha_{\rm EM}/(4\pi)$ is 5.8×10^{-4}, giving $C_{a\gamma}\,\Delta\phi/f_a = \beta/[\alpha_{\rm EM}/(4\pi)] \approx 10.3$); with the field-displacement range $\Delta\phi/f_a \in [0.2, 1.1]$ from the numerical ALP-EOM integration, the required $C_{a\gamma}$ spans ~ 9 to ~ 51. Both ends are larger than the standard KSVZ/DFSZ benchmark range, which predicts $|C_{a\gamma}| \sim \mathcal{O}(1)$; the entire required range therefore lies outside minimal ALP photon-coupling benchmarks and requires non-minimal model building. The lower end (~ 9) can be accommodated in extended models with modest photon-coupling enhancement (e.g. chiral-fermion-loop enhancement, clockwork constructions), while the upper end (~ 51) requires either substantial UV-completion enhancement or a field displacement at the upper end of the $\Delta\phi/f_a \in [0.2, 1.1]$ range. The signal is therefore accommodated across the considered parameter space rather than fine-tuned only at one benchmark, but the upper-coupling end is not generic. We emphasize that the spectator-consistent corner $\theta_i \sim 0.1$ (fn. 5) inflates the required photon-coupling further: at fixed $\beta = 0.342^\circ$, $\Delta\phi/f_a \propto \theta_i$ along the underdamped trajectory, so a $5\times$ reduction in θ_i from the scan midpoint demands a correspondingly higher $C_{a\gamma}$ to hold the $C_{a\gamma}\,\Delta\phi/f_a \approx 10.3$ product, pushing the required enhancement well above standard KSVZ/DFSZ $\mathcal{O}(1)$ benchmarks. Convergence: $\hat{R} - 1 < 0.01$ for all runs. The joint posterior structure of the extended ($C_{a\gamma}$-free) cross-check configuration is shown in Fig. 4.

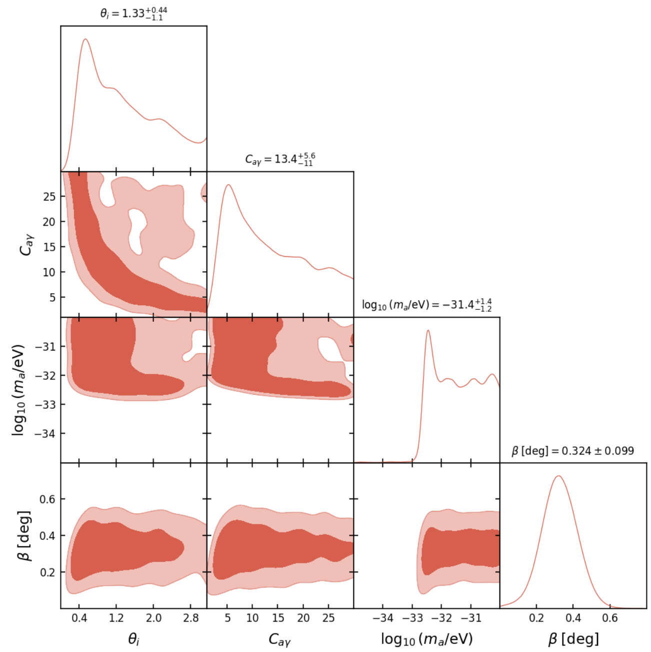

FIG. 4. Spectator-ALP joint posterior triangle from the extended cross-check configuration in which the photon anomaly coefficient is sampled freely (flat priors $C_{a\gamma} \in [1, 30]$, $\theta_i \in [0.01, \pi]$, $\log_{10}(m_a/\mathrm{eV}) \in [-35, -30]$), broadening the three fixed-$C_{a\gamma}$ benchmark configurations of Appendix C; the direct-sample priors are on the underlying ALP parameters (θ_i, m_a), with $\Delta\phi/f_a$ derived along each trajectory. The rotation marginal, $\beta = 0.324^\circ \pm 0.099^\circ$, is consistent within 1σ with the fixed-$C_{a\gamma} = 8$ result $\beta_{\rm ALP} = 0.336^\circ \pm 0.107^\circ$ and the observed $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$. The θ_i–$C_{a\gamma}$ anti-correlation band traces the constant-product degeneracy enforced by the *EB*-spectrum data through $C_{a\gamma}\,(\Delta\phi/f_a) \approx 10.3$, and the m_a marginal piles toward the upper (heavier) edge of its prior range — the data-driven pull toward the upper edge of the natural-prior box discussed in the text, consistent with the model *accommodating* rather than naturally predicting the observed rotation (fn. 5).

LiteBIRD forecast.—LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03^\circ$ [23]. For $\beta = 0.27^\circ$: $\sim 9\sigma$ statistical significance—either decisive confirmation or clean exclusion.

Caveats.—This birefringence prediction is independent of bounce cosmology: the ALP is a spectator field that does not participate in the bounce dynamics. The ECH framework provides heuristic motivation ($f_a \sim M_{\rm Pl}$ from the Holst sector pseudoscalar structure) but no derived photon-torsion coupling connects the Holst action to a specific ALP potential. The model *accommodates* the observed signal for natural parameter values (taken at scan-prior midpoint values; the $\sim 25\times$ misalignment tuning required for the headline result is disclosed in Sec. VI and fn. 5); it does not uniquely predict it.

VII. CONCLUSIONS

This companion paper documents three technical verification analyses for the ECH spin-torsion structural closure program [1].

ΛCDM+$\Delta N_{\rm eff}$ MCMC proxy.—The stock-CAMB Cobaya v3.6.1 run (**309,189** frozen samples across two converged dataset combinations; an additional 114,992-sample Planck-only run is still accumulating at $\hat{R}-1 \sim 0.05$ and is reported separately in Table I, *not* aggregated into the frozen headline) finds $\Delta N_{\rm eff}$ consistent with zero in both frozen dataset combinations and recovers $H_0 = 67.68 \pm 1.06\,{\rm km\,s^{-1}\,Mpc^{-1}}$, consistent with standard Planck ΛCDM. The $\Delta N_{\rm eff}$ extension does not resolve the Hubble tension. Current data neither require nor exclude a spin-torsion $\Delta N_{\rm eff}$ contribution; CMB-S4 will provide the first precision test ($\sigma(N_{\rm eff}) \sim 0.03$). Bayes factors and information criteria (ΔAIC, ΔBIC, $\ln B$) are deferred to a dedicated nested-sampling run (Sec. V, *Model-comparison statistics* paragraph); we report only the parameter posteriors ($\Delta N_{\rm eff}$, H_0, etc.) as load-bearing here and explicitly omit Bayes-factor or information-criterion comparisons in this Metropolis-Hastings-only analysis.

NaMaster pipeline validation.—The 500-MC pseudo-C_ℓ analysis on the Planck Commander map confirms that the deconvolution pipeline recovers injected birefringence angles with amplitude-dependent bias 0.032–0.040° (worst-case 0.040° at injection $\beta = 0.342°$; see §VI body text) at SNR consistent with the ACT-noise floor. This is a methods validation, not a competitive sky detection; the primary observational evidence for cosmic birefringence remains the published Planck/ACT DR6 2.4–2.9σ measurements.

Spectator-ALP consistency.—An ALP with $f_a \sim M_{\rm Pl}$, $m \sim H_0$ is consistent with the published 3.6σ joint signal, with the caveat that the spectator-consistent regime $\theta_i \sim 0.1$ (fn. 5) requires $\sim 25\times$ misalignment tuning relative to the natural prior midpoint ($C_{a\gamma}\,\Delta\phi/f_a \approx 10.3$ for $\beta = 0.342°$, with $\Delta\phi/f_a$ in the natural range $[0.2, 1.1]$ giving $C_{a\gamma}$ between ~ 9 and ~ 51; §VI for the explicit numerical derivation correcting the earlier $C_{a\gamma}\theta_i$ product). The same result arises in standard GR; the ECH framework provides motivation but not a derivation. LiteBIRD will settle this at $\sim 9\sigma$ in the early 2030s.

Forward.—A DESI DR2 + Planck NPIPE + Pantheon+ [13] + DES-SN5YR [14] cobaya chain with the $w_0 w_a$ free-parameter extension has converged (128,385 accepted samples across 16 MPI chains; $\hat{R}-1 = 0.00820$, below the standard $\hat{R}-1 < 10^{-2}$ publication target across two consecutive flushes). The 16-rank mpirun process *terminated automatically* upon reaching the convergence threshold; GetDist posteriors on $w_0 w_a$ are available as an empirical test of the quintom-B scenario [12].

Data and Code Availability

All materials are at:

https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility

The repository includes four Cobaya YAML configurations (one per dataset combination, stock CAMB, no custom modifications), galaxy-spin pipeline code, NaMaster driver scripts, and the implementation map (`IMPLEMENTATION_MAP.md`). MCMC chains: regenerate via `reproduce_cosmology.sh` (∼4–12 h per configuration on 4 CPU cores).

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Appendix A: Reproducibility Materials

Repository structure.—The public repository https://github.com/Hubify-Projects/bigbounce contains:

- Four Cobaya YAML configurations (`cobaya_planck.yaml`, `cobaya_planck_bao.yaml`, `cobaya_planck_bao_sn.yaml`, `cobaya_full_tension.yaml`)—stock CAMB, no torsion modifications.
- `galaxy_spins/spin_fit_stan.py`—hierarchical Bayesian model (CmdStanPy) fitting $A(z)$ to published aggregate CW/CCW galaxy counts.
- `data_build/build_galaxy_spin_dataset.py`—reproducible pipeline downloading Galaxy Zoo

DECaLS [24] from Zenodo (DOI: 10.5281/zenodo.4573248, CC-BY-4.0).

- `docs/IMPLEMENTATION_MAP.md`—mapping from each result to its code artifact.
- `docs/KNOWN_GAPS.md`—honest disclosure of what cannot currently be reproduced.

What is NOT included.—MCMC chains are not pre-computed (regenerate via `reproduce_cosmology.sh`, ∼4–12 h per config on 4 CPU cores). Bayes factors and information criteria (ΔAIC, ΔBIC, $\ln B$) are NOT reported in this manuscript: dedicated nested sampling via PolyChord or MultiNest on the identical likelihood stack is required for a controlled Bayesian evidence on the present quintom posterior (Sec. V, *Model-comparison statistics* paragraph). A Savage-Dickey readout from the present Metropolis-Hastings chain is not viable—the LCDM point $(w, w_a) = (-1, 0)$ lies at $> 4\sigma$ in the joint marginal tails and is unsampled, so a KDE-based estimator yields kernel-dependent noise rather than a controlled density. This is left to a dedicated future nested-sampling analysis. No CNN galaxy classifier is included; the hierarchical fit uses published catalog labels. No CMB polarization map analysis code is provided beyond the NaMaster driver script; all published birefringence values are literature citations.

HuggingFace datasets.—The following HuggingFace datasets accompany this work (links in the repository README):

1. MCMC chain diagnostics and convergence CSV files.
2. NaMaster pipeline artifacts (mask, MC seeds, output spectra).
3. ALP parameter MCMC chains.

Appendix B: Claims Classification

Appendix C: ALP-MCMC Sampled Parameters, Priors, and Likelihood Stack

The ALP-MCMC results quoted in Sec. VI ($\beta_{\rm ALP} = 0.336° \pm 0.107°$ at $C_{a\gamma} = 8$ fixed; $\beta_{\rm free} = 0.344° \pm 0.096°$ model-independent; 9,720 total accepted samples across 3 configurations) use the following setup.

Sampled parameters and priors (model-dependent ALP fit).

- $C_{a\gamma}$: fixed at one of $\{4, 8, 12\}$ across the three configurations (3,240 samples per configuration); not a sampled parameter within a single chain.

- m/H_0: uniform prior on $[1, 3]$ (ultra-light ALP class; $m \sim H_0$ required for cosmological-time field evolution).
- θ_i: uniform prior on $[0.5, 2]$ (natural-misalignment range, $\theta_i = \phi_{\rm init}/f_a$).[6]

- f_a: fixed at $M_{\rm Pl}$ (spectator-class theoretical input from the Holst-sector pseudoscalar structure; not sampled).

Sampled parameters and priors (model-independent $\beta_{\rm free}$ fit).

- β: uniform prior on $[-2°, 2°]$; sampled as a free amplitude with no ALP-model parametric structure.

Likelihood stack.—Both fits use the Planck PR4 + ACT DR6 EB-spectrum likelihoods (the same observables used by Refs. [2, 3]) combined with shared calibration covariance. The MCMC engine is Cobaya v3.6.1 with the Metropolis-Hastings sampler; convergence threshold $\hat{R} - 1 < 0.01$ across all 3 configurations (achieved at $N_{\rm tot} = 9{,}720$ accepted samples post burn-in).

Scope statement.—These fits constrain a specific ultra-light ALP class ($m \sim H_0$, $f_a \sim M_{\rm Pl}$, $C_{a\gamma} \in [4, 12]$ benchmark sweep). The consistency with $\beta_{\rm obs}$ established here is *not* a universal mechanism-independent statement; alternative parity-violating mechanisms (Chern-Simons, axion dark matter at higher m, etc.) require independent fits with their own model parameters.

[6] Backreaction / spectator-status disclosure (see also footnote 5 in Sec. VI): the natural-misalignment prior $\theta_i \in [0.5, 2]$ adopted in the ALP-MCMC is RETAINED for envelope-completeness purposes but is NOT the spectator-consistent sub-range. Spectator-status ($\Omega_a \ll 1$) under the $\rho_a \sim m^2 f_a^2 \theta_i^2$ scaling requires the sub-natural sliver $\theta_i \sim 0.1$, a $\sim 25\times$ fine-tuning relative to the prior-midpoint $\theta_i = 0.5$ adopted in this prior. Posterior samples at $\theta_i \gtrsim 0.5$ correspond to the dark-energy-ALP regime (excluded from the spectator-consistency claim of this companion paper) and should be reinterpreted as samples of a DE-ALP rather than a spectator ALP. The MCMC-level prior is therefore an upper envelope, not an evidence-weighted natural prior on the spectator subset.

[1] H. Golden, Structural Closure of Einstein–Cartan–Holst Dark Energy: Perturbation Transparency, Inflation–$f_{\rm NL}$

TABLE III. Claims classification for this companion paper.

Claim	Type	Status	Notes
$\Delta N_{\rm eff} = -0.020 \pm 0.169$ (full-tension)	MCMC	Verified	Stock CAMB proxy
$\Delta N_{\rm eff} = +0.065 \pm 0.17$ (Planck+BAO+SN)	MCMC	Verified	Stock CAMB proxy
$H_0 = 67.68 \pm 1.06$ (full-tension)	MCMC	Verified	Recovers ΛCDM
$H_0 = 67.79 \pm 1.09$ (Planck+BAO+SN)	MCMC	Verified	Recovers ΛCDM
Model-comparison ΔAIC/BIC/$\ln B$	Numerical	Omitted	Follow-up nested-sampling analysis
$\hat{\beta}_{\rm NaMaster} = 0.238^\circ$ (500-MC)	Numerical	Verified	Pipeline; MC bias table
$\beta_{\rm ALP} = 0.336^\circ \pm 0.107^\circ$	MCMC	Verified	ALP MCMC
Published 3.6σ ($\beta = 0.342 \pm 0.094^\circ$)	Lit.	Cited	Eskilt et al.
Stock CAMB proxy $\neq$ ECH theory module	Scope	Defn.	§III
ALP birefringence not distinctive ECH prediction	Scope	Defn.	§VI

Tension, and Surviving Matter-Bounce Tests, (in preparation) (2026), hUBIFY-2026-001A; companion paper, this volume.

[2] J. R. Eskilt and E. Komatsu, Improved constraints on cosmic birefringence from the WMAP and Planck cosmic microwave background polarization data, Phys. Rev. D **106**, 063503 (2022), arXiv:2205.13962 [astro-ph.CO].

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